

# **HYPERHEURISTIC OPTIMIZATION OF UNIVERSITY COURSE TIMETABLING PROBLEM USING REINFORCEMENT LEARNING AND GENETIC ALGORITHM**

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# Presentation Outline

- Motivation
- Introduction
- Objectives
- Literature Review
- Scope Of Project
- Project Applications
- Methodology
- Result
- Analysis
- Future Enhancements
- Conclusion
- References

# Motivation

- Motivation point 1: Describe the real-world problem or need
- Motivation point 2: Why this problem matters
- Motivation point 3: Current gaps or limitations
- Motivation point 4: What inspired this project

# Introduction

- Brief background of the project domain
- Key concepts and context
- Overview of the proposed system/solution

# Objectives

**Note:** Place the exact objectives as you submitted in the report. Remove this box before presenting.

- Objective 1: Primary goal of the project
- Objective 2: Secondary goal
- Objective 3: Additional goal
- Objective 4: Extended goal (if applicable)

# Literature Review

| Paper                               | Author          | Objective                                      | Methodology                            | Findings                               | Limitations                   |
|-------------------------------------|-----------------|------------------------------------------------|----------------------------------------|----------------------------------------|-------------------------------|
| Smart Traffic Signal Control (2021) | Sharma et al.   | Reduce urban congestion using adaptive signals | Reinforcement learning with Q-learning | 30% reduction in avg. wait time        | Tested only on simulated data |
| Crop Disease Detection (2022)       | Patel & Kumar   | Identify plant diseases from leaf images       | CNN with transfer learning (ResNet-50) | 94.5% accuracy on test set             | Limited to 5 crop species     |
| IoT-Based Air Quality (2023)        | Adhikari et al. | Real-time air quality monitoring system        | Sensor network with cloud dashboard    | Detected pollutant spikes within 2 min | High sensor calibration cost  |

# Scope Of Project

- Scope point 1: Define the boundaries and coverage of the project
- Scope point 2: What the project will deliver and its limitations

# Project Applications

- Application 1: Primary use case of the project
- Application 2: Secondary practical application
- Application 3: Broader impact or accessibility benefit



# Methodology - [1] (System Block Diagram)

**[System Block Diagram]**

Insert your system block diagram here

# Methodology - [2] (Working Principle)

- Working principle point 1: How the core system operates
- Working principle point 2: Sensor/data coordination
- Working principle point 3: Data capture mechanism
- Working principle point 4: Storage approach
- Working principle point 5: Data transmission and analysis

# Methodology - [3] (Operational Workflow)

- Workflow step 1: Signal preprocessing and filtering
- Workflow step 2: Noise/artifact handling
- Workflow step 3: Feature extraction from processed data
- Workflow step 4: Classification or decision-making
- Workflow step 5: Aggregation and final output

## Methodology - [4] (Flow Of Proposed System)

### **[System Flowchart]**

Insert your proposed system flowchart here

# Methodology - [5] (Hardware Requirement)

- Hardware component 1 (e.g. Microcontroller)
- Hardware component 2 (e.g. Sensor module)
- Hardware component 3 (e.g. Motion sensor)
- Hardware component 4 (e.g. Battery)
- Hardware component 5 (e.g. Power converter)

# Methodology - [6] (Software Requirement)

- Software tool 1 (e.g. IDE)
- Software tool 2 (e.g. Programming language)
- Software tool 3 (e.g. Web framework)
- Software tool 4 (e.g. ML library)
- Software tool 5 (e.g. Version control)
- Software tool 6 (e.g. Simulation tool)

# Methodology - [7] (Machine Learning Architecture)

- Description of the ML technique used
- Training approach and data splitting strategy
- How prediction/classification is performed
- Measures to prevent overfitting and improve generalization

# Methodology - [8] (Algorithm Architecture)

**[Algorithm Architecture Diagram]**

Insert your algorithm architecture diagram here



# Methodology - [9] (Dataset Preparation)

- Data source and collection method
- Key variables and measurements
- Labeling and annotation approach
- Feature extraction from raw data
- Train/validation/test split strategy

# Methodology - [10] (Snapshot of Dataset)

## **[Dataset Snapshot]**

Insert a sample table or screenshot of your dataset here

# Methodology - [11] (Model Training)

- Model/classifier training details
- Class balancing or weighting strategy
- Hyperparameter tuning and optimization metric
- Evaluation on held-out test set
- Model serialization and deployment approach

## Result[1] (Precision-Recall Curve)

**[Precision-Recall Curve]**

Insert your precision-recall curve figure here

## Result[2] (ROC Curve)

**[ROC Curve]**

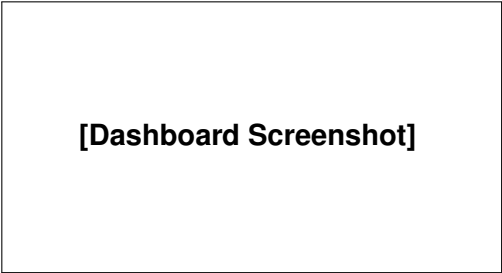
Insert your ROC curve figure here

## Result[3] (Confusion Matrix)

- Classification performance summary
- True positives and false negatives
- True negatives and false positives
- Accuracy, Precision, Recall metrics

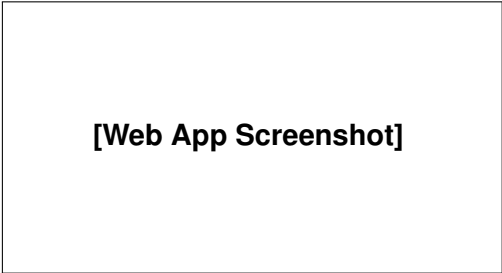
**[Confusion Matrix]**

## Result[4]



**[Dashboard Screenshot]**

- Dashboard for data visualization



**[Web App Screenshot]**

- Homepage for web app

## Result[5]

**[Prototype Photo]**

Final prototype of the system

**[Data Table]**

Data received from the hardware



# Analysis

## Data Analysis:

- Analysis point 1: Key observation from the data
- Analysis point 2: Consistency and reliability of measurements
- Analysis point 3: Statistical summary of results
- Analysis point 4: Comparison with expected values
- Analysis point 5: Data logging and storage observations

# Future Enhancements

- Future enhancement 1: Broader dataset and improved generalization
- Future enhancement 2: Advanced model architecture for better performance
- Future enhancement 3: Edge deployment for real-time operation

# Conclusion

- Conclusion point 1: Summary of the proposed/developed system
- Conclusion point 2: Key contributions and significance
- Conclusion point 3: Feasibility and approach taken
- Conclusion point 4: Results achieved or expected outcomes
- Conclusion point 5: Practical impact and relevance

# References

- Reference 1: Author(s), “Title of paper/article,” Journal/Conference, Date.  
Available: URL. [Accessed: Date]
- Reference 2: Author(s), “Title of paper/article,” Journal/Conference, vol. X, no. Y, pp. XX–YY, Year

## References(Contd..)

- Reference 3: Author(s), "Title of paper/article," Journal/Conference, vol. X, p. XXXXX, Month. Year
- Reference 4: Author(s), "Title of paper/article," Journal/Conference, vol. X, no. Y, pp. XX–YY, Year